

Binary Batch Distillation

Final Lab Report
Unit Operations Lab 2
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Group #2
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1. Executive Summary (10 pts)

The objective of the lab was to determine the boil-up rates, top/bottom composition, pressure drop, and column efficiency by altering reboiler power rates and reflux ratios. A solution containing 10 wt% ethanol and 90 wt% water was created and poured into the reboiler. Using the control panel, the temperature and power were adjusted until reflux was achieved. The distillate was collected from the column, and three samples from the reboiler were placed into a graduated cylinder and timed volumetrically. The samples were placed into test tubes and had their temperatures and concentrations measured with a Snap 41 Densitometer. The process was repeated three times for Experiment B, with power set to 0.75 kW and reflux ratios of 5:1 and 12:1.

Experiment A started with 0.75 kW, increasing the power by 0.25 kW until 1.50 kW. The results were: 77 Δp , 1.91 L/hr; 80 Δp , 3.09 L/hr; 114 Δp , 3.86 L/hr; 168 Δp , 9.41 L/hr. For Experiment B, the average results at 5:1 were 2.94 L/hr, 84.61 mol% ethanol distillate, and 59.37 mol% ethanol bottoms. At 12:1, they were 1.53 L/hr, 93.04 mol% ethanol distillate, and 24.64 mol% ethanol bottoms. This demonstrates the compromises made with binary batch distillations: higher power increases the boil rate but also increases the pressure drop, while higher reflux ratios improve efficiency; however, productivity is reduced.

2. Introduction (10 pts)

A solution consisting of 10 wt% alcohol and 90 wt% water was fed into a distillation column. The purpose of the binary batch distillation was to investigate how temperature affects the alcohol composition in both the distillate and the reboiler. A second experiment was conducted using the same column to examine the effect of varying the reflux ratio on the distillate composition.

According to the lab manual (Armfield Limited, 2014), the reboiler was initially charged with 10 L of the alcohol–water solution. Power was then supplied, and the column was operated until steam reached the condenser at the top. Cooling water was introduced to condense the vapor into distillate. Once steady-state conditions were established, indicated by a constant temperature, samples were collected from both the distillate and the reboiler using graduated cylinders. The collection time was recorded. Portions of the distillate were transferred to test tubes, chilled in ice water, and subsequently analyzed along with the reboiler samples using the Snap 41 to determine alcohol concentration. This procedure was repeated for three additional power levels. Because the unit operated as a closed system with recycle, no solution was lost; distillate was continuously returned to the reboiler.

For the second part of the experiment, the power level was adjusted to the required setting, and the column was operated until equilibrium was reached. At equilibrium, the

operating temperature was recorded, and the desired reflux ratio was applied. Samples were collected from valves 2 and 3, and their temperatures were measured. The procedure was repeated at different reflux ratios, and the distillate compositions were determined using Snap 41.

The purpose of the lab was to demonstrate how operating at different power levels and reflux ratios influences the alcohol composition in both the distillate and the reboiler (McCabe, Smith, & Harriot, 2005). This understanding is relevant to industrial applications, where distillation efficiency and purity are critical. For example, in pharmaceutical production of medical-grade alcohol, high-purity alcohol is required for sanitation, which demands significant energy input to achieve separation of alcohol from water (Nedstar B.V., 2025). Another application is in crude oil refining, where different temperatures produce different fractions, and unseparated material is recycled through the separator to maximize yield (U.S. Energy Information Administration, 2025).

3. Theory (10 pts)

Distillation is a significant separation technique in chemical engineering due to its ability to produce highly purified products by applying principles of phase equilibrium. When considering a binary batch distillation system, two compounds of distinct volatilities like ethanol and water are divided as vapor ascends and liquid descends through trays. Even if it would be desirable to have each tray operate at vapor–liquid equilibrium, this is impractical; thus, an introduction to column efficiency is provided.

The most significant connection is that when reboiler power increases, the rate of vapor boil-up is greater. This results in a higher velocity of vapor through the trays, which in turn leads to a higher pressure drop along the column. Although higher rates of boil-up yield higher mass transfer rates, they can result in foaming and entrainment that reduce overall effectiveness.

The column efficiency is defined as a comparison of the actual separation to that of an ideal separation as predicted by a perfect column. It is possible to estimate the number of theoretical stages from distillate and bottoms compositions by relative volatility and equilibrium data. Efficiency is calculated by dividing the theoretical plates by the actual plates and then multiplying by 100%. Industrial-grade distillation columns typically reach 70–90% efficiency, whereas lab columns are expected to operate at lower efficiencies due to heat loss, scale restrictions, and incomplete phase contact.

Another critical parameter is the reflux ratio. An increased reflux ratio sends a higher volume of condensed distillate to the column, thereby enhancing separation and improving the purity of the distillate. This is a compromise between purity and productivity, as it decreases distillate throughput per unit time. These associations provide the basis of a theoretical experiment to be carried out.

4. Results (15 pts)

Influence of Reboiler Power on Boil-up Rate and Pressure Drop

The experiment validated that an increase in reboiler power resulted in higher rates of boil-up and a greater pressure drop along the column. Recorded results are given below:

For 0.75 kW: $\Delta p = 77$, boil-up rate = 1.91 L

At 1.00 kW: $\Delta p = 80$, boil-up rate = 3.09 L

At 1.25 kW: $\Delta p = 114$, boil-up rate = 3.86 L

At 1.50 kW: $\Delta p = 168$, boil-up rate = 9.41 L

This trend conformed to theoretical prediction. As vapor velocity increased proportionally to power, stability decreased, resulting in foaming on trays at higher powers.

Effect of Reflux Ratio on Distillate and Bottom Compositions:

At a constant power input of 0.75 kW, the reflux ratio varied. These results revealed that lower productivity resulted from higher reflux ratios, but higher ethanol purity in distillate resulted from higher reflux ratios:

At 5:1 reflux: 2.94 L/hr boil-up rate, distillate = 84.61 mol% ethanol, bottoms = 59.37 mol

At 12:1 reflux: Boil-up rate = 1.53 L/hr, distillate = 93.04 mol% ethanol, bottoms = 24.64 mol% ethanol

These results demonstrate the anticipated compromise: increasing reflux results in higher purity but lower overall recovery rate. Tabulated in Figures 1–7 (Presented in Appendix I) are the experimentally measured pressure drops, boil-up rates, and efficiencies.

5. Discussion (15 pts)

Rate of Boil-up and Efficiency:

As reboiler power increased from 0.75 to 1.50 kW, the pressure drop of the column and the boil-up rate of liquid into trays increased. This is consistent with the manual explanation that pressure drop is a result of the velocity of vapour through trays.

Efficiency and VLE plots of higher powers (1.25–1.50 kW) indicate that ethanol recovery improved due to intensified vapor flow; however, the improvement in efficiency began to decline as liquid foaming increased.

Column Efficiency:

From the Fenske equation and the Murphree definition of efficiency. Efficiencies derived from those were below optimal. Industrial columns typically operate at 70–90% efficiency, but our lab results were lower than this due to non-ideal lab-scale considerations, including tray wetting, foaming, and exhaust losses.

Reflux Ratio:

The variation in reflux ratio caused considerable fluctuations in top and bottom ethanol concentrations over time. As would be expected, stronger reflux led to a superior separation higher purity of ethanol in the distillate, but a lower distillate yield per unit time. This is an industrial compromise between the input of energy and product purity. In short, empirical results validated the theoretical connection between separation efficiency and boil-up rate as well as reflux ratio. These deviations from literature

correlations could be due to scaling constraints, sub-optimality of tray equilibrium, and heat losses.

6. References (10 pts)

1. Armfield Limited, UOP3CC Continuous Distillation Column Instruction Manual, Issue 38 March 2014.
2. W. McCabe, J. Smith, and P. Harriot, Unit Operations of Chemical Engineering, 7th ed. McGraw-Hill, 2005.
3. Nedstar B.V. (n.d.). *Pharmaceutical grade alcohol*, 2025.
4. U.S. Energy Information Administration. n.d. *Refining Crude Oil: The Refining Process*, 2025.
5. P. C. Wankat, Equilibrium-Staged Separations. Prentice Hall, 1988
6. Kunesh, H. Kister, and et al., "Distillation: Still towering over other options," Chemical Engineering Progress, vol. 89, p. 43, 1995.

7. Appendix I: Data Tabulation/Graphs (10 pts)

PART A:

Part A		
Power (kW)	ΔP	Boil-up Rate (L/hr)
0.75	77	1.91
1.00	80	3.09
1.25	114	3.86
1.50	168	9.41

Figure 1 – Pressure Drop & Boil-up Rate at different boiler power

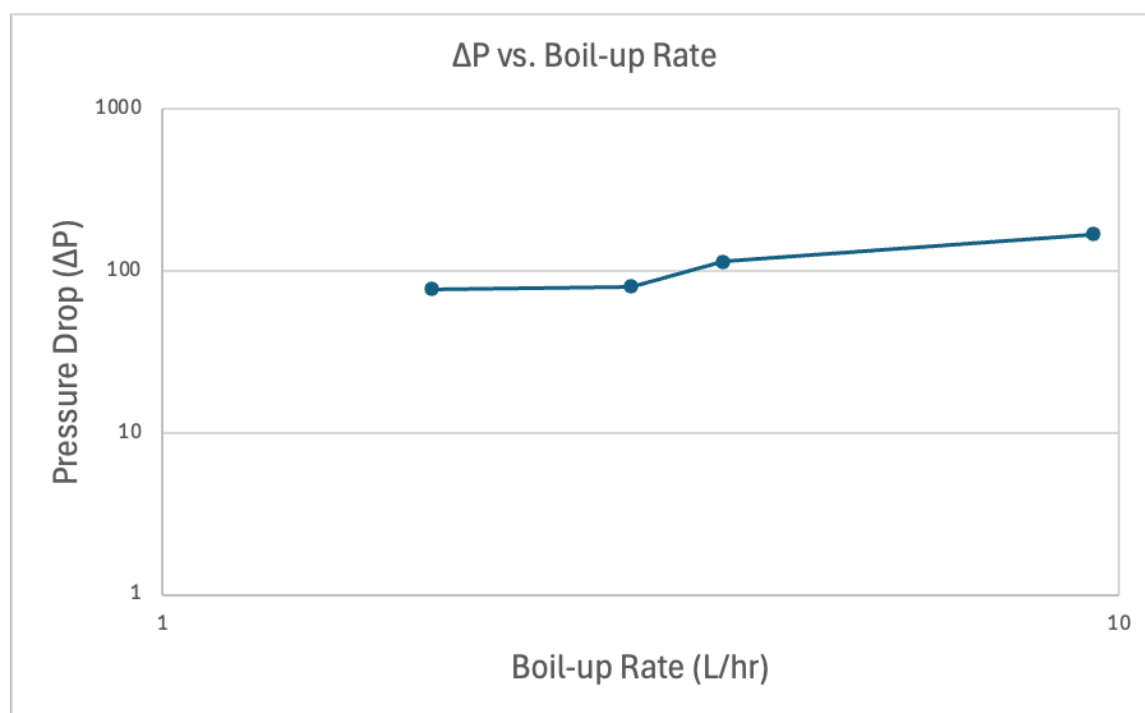


Figure 2 - Pressure Drop vs. Boil-up Rate

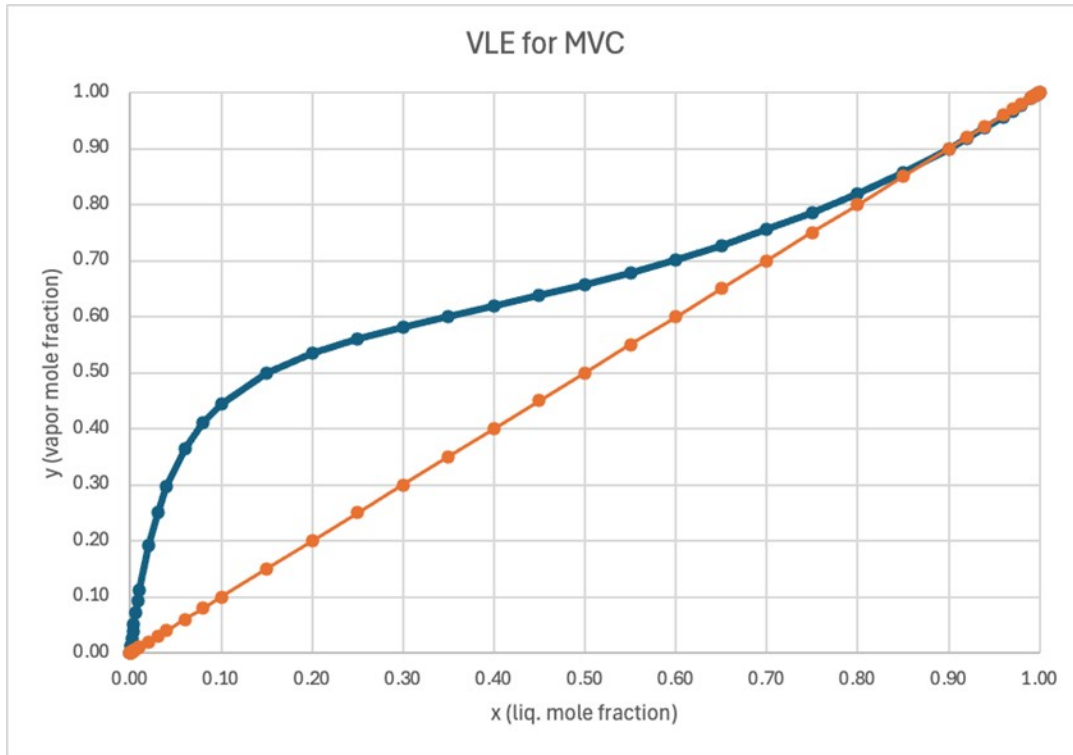


Figure 3 - VLE Data for MVC (EtOH)

	1.00 kW		
X_etoh_D	0.9839	y_etoh_D	0.9780
X_etoh_B	0.1351	Y_etoh_B	0.1281
X_h20_D	0.0161	Y_h20_D	0.0220
Xh20_B	0.8649	Y_h20_B	0.8719
$\alpha_{\text{etoh/h20_D}}$	0.7274		
$\alpha_{\text{etoh/h20_B}}$	0.9406		
$\alpha_{\text{etoh/h20_avg}}$	0.8272		
D ratio	61.1118		
B ratio	6.4019		
n	32.4586		
Efficiency	360.6507	%	

Figure 4 – Efficiency Data at 1.00 kW

	1.25 KW		
X_etoh_D	0.9860	y_etoh_D	0.9772
X_etoh_B	0.2420	Y_etoh_B	0.5332
X_h2o_D	0.0140	Y_h2o_D	0.0228
Xh2o_B	0.7580	Y_h2o_B	0.4668
$\alpha_{\text{etoh/h2o_D}}$	0.6086		
$\alpha_{\text{etoh/h2o_B}}$	3.5778		
$\alpha_{\text{etoh/h2o_avg}}$	1.4756		
D ratio	70.4286		
B ratio	3.1322		
n	12.8710		
Efficiency	143.0116	%	

Figure 5 – Efficiency Data at 1.25 kW

	1.50 KW		
X_etoh_D	0.9903	y_etoh_D	0.9888
X_etoh_B	0.4038	Y_etoh_B	0.6192
X_h2o_D	0.0097	Y_h2o_D	0.0112
Xh2o_B	0.5962	Y_h2o_B	0.3808
$\alpha_{\text{etoh/h2o_D}}$	0.8648		
$\alpha_{\text{etoh/h2o_B}}$	2.4008		
$\alpha_{\text{etoh/h2o_avg}}$	1.4409		
D ratio	102.0928		
B ratio	1.4765		
n	12.7317		
Efficiency	141.4630	%	

Figure 6 – Efficiency Data at 1.50 kW

PART B:

				5:1 Ratio		(Power = 0.75 kW)			
Trial	Volume (mL)	Time (s)	Boil-up Rate (L/hr)	X_EtOH_D (mol %)	X_EtOH_B (mol %)	T1 (deg C)	T8 (deg C)	Avg. Temp (deg C)	
1	10.00	12.26	2.94	82.27	59.48	78.70	80.60	79.65	
2	10.00	11.97	3.01	89.31	57.42				
3	10.00	12.57	2.86	82.24	61.22				
Average =			2.94	84.61	59.37				
				12:1 Ratio		(Power = 0.75 kW)			
Trial	Volume (mL)	Time (s)	Boil-up Rate (L/hr)	X_EtOH_D (mol %)	X_EtOH_B (mol %)	T1 (deg C)	T8 (deg C)	Avg. Temp (deg C)	
1	10.00	23.86	1.51	93.20	23.41	78.60	83.10	80.85	
2	10.00	24.89	1.45	91.40	27.54				
3	10.00	22.19	1.62	94.51	22.98				
Average =			1.53	93.04	24.64				

Figure 7 – Part B Data

8. Appendix II: Error Analysis (10 pts)

Instrumental Errors:

The Snap 41 densitometer calibration would have introduced a ± 0.001 – 0.002 g/mL error in density, equivalent to a 1–2 mol% uncertainty in ethanol. The boil-up rate stop-watch and cylinder timing involved a ± 2 sec error, producing a 3–5% uncertainty when measurements of L/hr were made.

Systematic errors:

Dissipation of heat to the surroundings reduced its adequate reboiler capacity, thus lowering efficiency compared to theoretical expectations. Incomplete condensation of higher boil-up rates might lead to losses of ethanol vapor.

Stochastic Errors: Foaming and entrainment, which occurred when higher powers were applied, resulted in inconsistent pressure drop measurements due to sampling error (loss during evaporation when transporting warm bottom samples).

Error Propagation: The propagated uncertainty of the calculated efficiency was ± 7 – 10% with $\pm 5\%$ error in composition and $\pm 3\%$ in boil-up rate.

Influence on Conclusions: Even so, overall trends (boil-up rate $\uparrow \rightarrow \Delta P \uparrow$; reflux $\uparrow \rightarrow$ distillate purity $\uparrow$) held.

9. Appendix III: Sample Calculations (10 pts)

ChemE Lab Cesearo D.
29 Sept 2025

Distillation [Sample Calculations]

Power at 0.75 kW

Avg. Boil-up : $1.91 \text{ L} \cdot \text{hr}^{-1}$
Rate

Distillate EtOH : (a) 92.6 mol %	Distillate H ₂ O : (a) 7.4 mol %
<u>Composition</u>	<u>Composition</u>
(b) 93.01 mol %	(b) 6.99 mol %
(c) 91.02 mol %	(c) 8.98 mol %

Avg. EtOH : 92.21 mol %	Avg. H ₂ O : 7.79 mol %
<u>composition</u>	<u>Composition</u>

Bottoms EtOH : (a) 84.62 mol %	Bottoms H ₂ O : (a) 15.38 mol %
<u>composition</u>	<u>Composition</u>
(b) 83.76 mol %	(b) 16.24 mol %
(c) 84.12 mol %	(c) 15.88 mol %

Avg. Bottoms EtOH : 84.17 mol %	Avg. Bottoms H ₂ O : 15.83 mol %
<u>composition</u>	<u>composition</u>

Avg. T₁ : 78.8 °C

Avg. T₈ : 81.1 °C

Avg. Column Temp. : 79.95 °C

Part A – Efficiency Calculations At 0.75 kW Boiler Power

of Theoretical Plates (n)

$$|n| = \frac{\log \left[\left(\frac{x_{\text{EtOH}}}{x_{\text{H}_2\text{O}}} \right)_D \left(\frac{x_{\text{H}_2\text{O}}}{x_{\text{EtOH}}} \right)_B \right]}{\log(\alpha_{\text{EtOH}/\text{H}_2\text{O}})_{\text{avg}}} - 1 = 22.3$$

$-(x_{\text{EtOH}})_D = 0.9221$ $-(y_{\text{EtOH}})_D = 0.9180$
 $-(x_{\text{EtOH}})_B = 0.8417$ $-(y_{\text{EtOH}})_B = 0.8391$
 $-(x_{\text{H}_2\text{O}})_D = 0.0779$ $-(y_{\text{H}_2\text{O}})_D = 0.0820$
 $-(x_{\text{H}_2\text{O}})_B = 0.1583$ $-(y_{\text{H}_2\text{O}})_B = 0.1609$

$$-(\alpha_{\text{EtOH}/\text{H}_2\text{O}})_D = \frac{(K_{\text{EtOH}})_D}{(K_{\text{H}_2\text{O}})_D} = \frac{(y_{\text{EtOH}}/x_{\text{EtOH}})_D}{(y_{\text{H}_2\text{O}}/x_{\text{H}_2\text{O}})_D} = 0.9460$$

$$-(\alpha_{\text{EtOH}/\text{H}_2\text{O}})_B = \frac{(K_{\text{EtOH}})_B}{(K_{\text{H}_2\text{O}})_B} = \frac{(y_{\text{EtOH}}/x_{\text{EtOH}})_B}{(y_{\text{H}_2\text{O}}/x_{\text{H}_2\text{O}})_B} = 0.9808$$

$$-(\alpha_{\text{EtOH}/\text{H}_2\text{O}})_{\text{avg}} = \sqrt{(\alpha_{\text{EtOH}/\text{H}_2\text{O}})_D \times (\alpha_{\text{EtOH}/\text{H}_2\text{O}})_B} = 0.9631$$

$$\text{Efficiency} = \frac{22.3}{9} \times 100 = 247.8\%$$

Part A - Efficiency Calculations At 0.75 kW Boiler Power (cont.)

10. Appendix IV: Individual Team Contributions

Team Contributions

	TIME (HOURS)	Team Member(s)
Operator (Both Lab Days)	4 hours	Ameen Abuzir,Joshua Laforest,Javier Silva, Cesearo Dilosa
Pre-Lab Editing	1 hour	Ameen Abuzir,Joshua Laforest,Javier Silva, Cesearo Dilosa
Experimental Objective (Prelab)	30 minutes	Joshua Laforest
Apparatus (prelab)	30 minutes	Joshua Laforest
Materials & Supplies (prelab)	30 minutes	Joshua Laforest
Procedure (prelab)	30 min	Javier
Project Safety Evaluation (prelab)	30 min	Javier
Final Lab Editing	2 Hours	Ameen Abuzir,Joshua Laforest,Javier Silva, Cesearo Dilosa
Executive Summary (final lab)	40 minutes	Joshua Laforest
Introduction (Final lab)	45 minutes	Javier
Theory (Final Lab)	1 hour	Ameen
Results (final lab)	30 minutes	Ameen
Discussion (final lab)	45 minutes	Ameen
References (final and/or prelab)	20 min	Javier
Data Tabulation/Graphs (final)	2 hours	Cesearo
Error Analysis (final lab)	45 minutes	Ameen
Sample Calculations (final Lab)	1 Hour	
Power Point Presentation	1 Hour	Ameen Abuzir,Joshua Laforest,Javier Silva,

		Cesearo Dilosa
Total	18 hours 15 minutes	